

ASME B31.3 Pipe Wall Thickness Report

Report Generated: 2026-06-11 10:29:05

Input Parameters

Nominal Pipe Size	NPS 6
Outside Diameter	168.3 mm
Design Pressure	1.034 MPa
Design Temperature	260 °C
Material	A53 Grade B
Weld Type	Seamless
Material Type	Ferritic
Corrosion Allowance	1.5875 mm
Threaded Connection	Yes
Mill Tolerance	12.5%

Calculation Details

Pressure Design Thickness (t _{dis})	0.666 mm
Corrosion Allowance (CA)	1.587 mm
Thread Depth (TD)	2.540 mm
Allowable Stress (S)	130.31 MPa
Weld Joint Efficiency (E)	1.000
Y Coefficient (Y)	0.400

Results Summary

Required Thickness (t _{req})	4.793 mm
Minimum Thickness (t _{min})	5.478 mm
Thin-Wall Condition	OK
Overall Status	PASS

ASME B36.10M Schedule Thicknesses

Schedule	Thickness (mm)	Status
5	3.40	✗ Inadequate
10	5.54	✓ Adequate
40	7.11	✓ Adequate
80	10.31	✓ Adequate
120	12.70	✓ Adequate

160	15.09	✓ Adequate
XXS	17.45	✓ Adequate